Analyzing Facebook Ego Network Topology and Identifying Communities

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Abstract

Online social networks have become one of the most important sources of real-world network data. Platforms such as Facebook make it possible to observe how people build friendships, form social circles, and share information at a very large scale. In this project, we analyze Facebook ego networks using ideas from complex network theory. We represent users as nodes and friendships as edges, and then study structural properties such as degree distribution, density, clustering, shortest-path behavior, centrality, modularity, structural holes, and diffusion. We also compare the Facebook network with Barabási–Albert and Erdős–Rényi synthetic models in order to better understand how real social networks differ from generated graph models. Community detection is performed using the Louvain method, while information spreading is studied using the Independent Cascade model. Our results show that the Facebook network is globally sparse but locally dense, follows a heavy-tailed degree pattern, contains strong and interpretable communities, and supports quick but incomplete information spread because of modular boundaries. These results suggest that Facebook ego networks have both small-world and scale-free characteristics. Overall, this project shows that graph-based analysis can reveal hidden patterns in online social structure and can help identify influential users, dense friendship groups, and bridge nodes that connect different communities.

1 Introduction

Facebook, Instagram, LinkedIn, and X are examples of online social networks (OSNs) that have changed how people talk to each other and stay in touch. People can use these platforms to talk to friends, family, classmates, coworkers, and other people who share their interests. They also create a lot of relational data at the same time. Network science can use these relationships to make graphs, where users are nodes and social ties are edges. This is very useful.

Using graphs to analyze data lets us look into a lot of interesting questions. We can, for instance, ask if a network is very clustered, if it has a few very important users, if it has well-defined communities, or if information can move quickly across the whole graph. These questions are important not just in theory, but also in real life, like when it comes to social recommendation systems, viral marketing, tracking false information, and campaigns to raise public awareness.

The *ego network* is a very useful idea in this field. An ego network looks at one main user, called the ego, and the people who are directly connected to that user, called alters. It also looks at the links between those alters. Ego networks are easier to read than the full global graph, but they still have a lot of useful social data. They can show strong friend groups, bridge users, local clustering, and how a person's social world is set up. Past research has demonstrated a significant correlation between ego networks and social circles, community structure, and the dissemination of information online. [1, 2, 10, 11].

In this project, we analyze Facebook ego networks using methods from complex network analysis. The work follows the main ideas of the original project proposal and focuses on graph topology, centrality, clustering, community detection, structural holes, k-core ideas, and diffusion. Instead of relying on only one graph metric, we combine several methods so that we can get a more complete understanding of the network.

1.1 Problem Statement

Facebook friendship networks are large, irregular, and highly structured. Because of this, it is difficult to understand them using only simple statistics such as the total number of users or the average number of friends. Important tasks such as identifying influential users, finding hidden communities, measuring clustering, and modeling information spread require specialized graph-analysis techniques. Also, importance in a social network is not one-dimensional. A user with many friends may not be the same as a user who connects multiple groups. In some cases, a node with fewer total connections may have greater strategic importance because it acts as a bridge between otherwise separate communities.

The main question addressed in this project is:

> How can complex network theory be used to analyze Facebook ego networks and reveal important structural properties such as clustering, centrality, community organization, and diffusion behavior?

1.2 Project Objectives

The main objectives of this paper are:

(1) Model the Facebook ego network as an undirected graph.
(2) Compute global and local topological properties such as degree, density, clustering, average path length, and diameter.
(3) Use centrality measures to identify important users.
(4) Detect communities using Louvain modularity optimization.
(5) Interpret bridge nodes through structural-hole ideas.
(6) Examine the core-periphery structure using k-core decomposition.
(7) Simulate information spread using the Independent Cascade model.
(8) Compare the Facebook graph with Barabási–Albert and Erdős–Rényi synthetic models.

1.3 Why This Project Matters

This topic matters because understanding network structure can be useful in many practical settings. Businesses can use it to identify influential users for promotion or advertising. Researchers can use it to study how opinions, trends, or rumors spread. Universities and organizations can use it to understand collaboration groups or peer structures. In short, network analysis helps answer questions such as:

- Which users are the most central in the network?
- Which users act as bridges between groups?
- How many meaningful communities exist?
- How tightly connected are local friendship circles?
- How quickly can information spread?

1.4 Paper Organization

The rest of this paper is organized as follows. Section 2 reviews the most relevant literature. Section 3 describes the dataset and preprocessing steps. Section 4 presents the mathematical framework. Section 5 explains the methodology. Section 6 presents the results. Section 7 discusses the findings in a more human and social context. Section 8 describes project limitations and future work. Section 9 concludes the paper.

2 Literature Review

The analysis of online social networks utilizes graph theory, sociology, statistical physics, and data mining. This part gives a brief overview of the main ideas and papers that helped us with our project.

2.1 Scale-Free Networks and Preferential Attachment

Barabási and Albert made one of the most important discoveries in network science when they showed that many real-world networks don't act like random graphs. [4]. Many real networks have heavy-tailed degree distributions instead of narrow ones. This means that most nodes have only a few connections, but a few nodes have very high degrees. They used preferential attachment to explain this pattern, which means that new nodes are more likely to connect to nodes that are already well-connected. This idea is very important for online social networks because some users naturally become hubs over time.

2.2 Small-World Behavior in Social Graphs

People also call social networks *small-world* networks. This means that they have short paths and high clustering. Even if a graph has thousands or millions of users, it can still be surprisingly easy to get from one user to another. Ugander et al. examined the anatomy of the Facebook social graph and found significant clustering and minimal distances between users [14]. Backstrom et al. also talked about the idea of "four degrees of separation" on Facebook, which showed that users are socially much closer than most people think [3]. These studies back up our use of metrics for average path length, diameter, and clustering.

2.3 Ego Networks and Social Circles

The concept of ego networks is especially useful in social-network analysis because it provides a local but meaningful view of the graph. Arnaboldi et al. studied ego network structure in online social networks and found that many online patterns still reflect offline human social behavior [2]. In later work, they also showed that ego networks play a major role in online information diffusion [1].

McAuley and Leskovec studied how to discover social circles inside ego networks and showed that community-like structures can often correspond to real-life groups such as family, school friends, or colleagues [11]. Leskovec and McAuley later discussed ego networks more broadly and explained why they are useful in social-network analysis [10].These studies strongly motivate our emphasis on ego-level structure and community detection.

2.4 Community Detection and Modularity

Discovering communities is one of the most essential tasks to do in network science. Clauset, Newman, and Moore suggested important ways to find community structure in big networks [6]. Newman subsequently formalized modularity as an indicator of the robustness of a network partition [12]. Modularity looks at how many edges there are in the real network that connect nodes in the same community and how many edges there should be in a null model.

Blondel et al. introduced the Louvain method, which is both fast and useful for optimizing modularity [5]. Other pertinent research encompasses label propagation by Raghavan et al. [13], consensus clustering by Lancichinetti and Fortunato [9], and Fortunato's comprehensive review of community detection methodologies [7]. Yang and Leskovec also looked at how detected communities stack up against real groups [15].

2.5 Homophily, Clustering, and Latent Structure

In social networks, users often connect to others who are similar to them. This is called homophily. It can be based on education, location, work, age, or interests. Homophily naturally leads to clustering and community formation. Krivitsky et al. studied how degree structure, clustering, and homophily can be modeled using latent random-effects approaches [8]. Their work supports the idea that dense local neighborhoods and triadic closure are not accidental, but natural social outcomes.

2.6 Literature Summary

The literature suggests that Facebook-like networks should show:

- heavy-tailed degree distributions,
- high local clustering,
- short average path lengths,
- clear community structure,
- and diffusion patterns shaped by bridge nodes and weak ties.

Our project tests these expectations on the Facebook ego network dataset by combining several graph measures and visualizations.

3 Dataset Description and Preprocessing

3.1 Dataset Source

The main dataset used in this project is the Facebook ego network dataset from the Stanford Large Network Dataset Collection (SNAP) [11]. The project proposal also identified the Network Repository and the TIB Data Repository as related sources, but this report focuses on the SNAP dataset because it is the standard benchmark for this kind of analysis and aligns directly with the reference papers listed in the proposal.

3.2 Dataset Characteristics

The combined Facebook graph used in our analysis contains:

- 4,039 nodes,
- 88,234 edges,
- an undirected friendship structure,
- anonymized user identities.

These values are large enough to show meaningful social structure but still manageable for graph analysis and visualization in a course project setting.

3.3 Preprocessing Steps

The preprocessing process used in this project included:

(1) Reading the edge list into Python.
(2) Constructing an undirected graph.
(3) Removing self-loops and duplicate edges if present.
(4) Making sure that the components are linked and using the main connected component for path-based analysis.
(5) Taking out certain ego-network ideas to give them a more in-depth analysis.

3.4 Why Preprocessing Is Important

Preprocessing matters because graph results can be affected by invalid edges, disconnected components, or duplicate records. For example, average path length is easier to interpret on a connected component, and self-loops do not represent meaningful Facebook friendships in this context. By cleaning the data first, the later analysis becomes more reliable.

4 Mathematical Framework

4.1 Graph Representation

We represent the Facebook network as an undirected graph $G = (V, E)$, where V is the set of users and E is the set of friendship connections. If the graph has $n = |V|$ nodes and $m = |E|$ edges, then the adjacency matrix A is defined as:

$$A_{ij} = \begin{cases} 1, & \text{if nodes } i \text{ and } j \text{ are connected,} \\ 0, & \text{otherwise.} \end{cases}$$

4.2 Degree and Density

The degree of node i is:

$$k_i = \sum_j A_{ij}.$$

The average degree of the network is:

$$\langle k \rangle = \frac{2m}{n}.$$

The density of the graph is:

$$D = \frac{2m}{n(n-1)}.$$

Density tells us how full the graph is compared to a complete graph. In most social networks, the density is low because users cannot be connected to everyone.

4.3 Shortest Paths

If $d(i, j)$ is the shortest-path distance between nodes i and j, the average path length is:

$$L = \frac{1}{n(n-1)} \sum_{i \neq j} d(i, j).$$

The diameter is the largest shortest-path distance between any two nodes in the network. These measures help explain how easily information can travel.

4.4 Clustering Coefficient

For a node i with degree k_i, the local clustering coefficient is:

$$C_i = \frac{2e_i}{k_i(k_i - 1)},$$

where e_i is the number of links among the neighbors of node i. This captures triadic closure, which is the idea that friends of friends often become friends.

4.5 Centrality Measures

We use four related ideas of node importance.

Degree Centrality. A node with high degree has many direct connections and can be seen as locally popular.

Betweenness Centrality.

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}},$$

where σ_{st} is the total number of shortest paths between nodes s and t, and $\sigma_{st}(v)$ is the number of those paths passing through v.

Closeness Centrality.

$$C_C(v) = \frac{n-1}{\sum_{t \in V} d(v, t)}.$$

Eigenvector Centrality.

$$x_v = \frac{1}{\lambda} \sum_{u \in V} A_{vu} x_u.$$

These measures highlight different kinds of importance. A user can be popular, globally reachable, connected to other important users, or strategically important as a bridge.

4.6 Modularity

If nodes are assigned to communities, the modularity score is:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j),$$

where c_i is the community label of node i and $\delta(c_i, c_j) = 1$ if nodes i and j are in the same community. A high modularity score indicates strong community structure.

4.7 Structural Holes

Bridge users can be interpreted through structural-hole theory. A node that connects different communities may hold a brokerage advantage. A related measure is constraint:

$$c_{ij} = \left(p_{ij} + \sum_{q \neq i,j} p_{iq} p_{qj} \right)^2, \qquad C_i = \sum_j c_{ij}.$$

Lower constraint usually suggests stronger brokerage potential.

4.8 k-Core Decomposition

The k-core of a graph is the maximal subgraph in which every node has degree at least k inside that subgraph. This helps separate the dense, stable part of the network from the weaker outer periphery.

4.9 Independent Cascade Model

In the Independent Cascade model, each newly activated node gets one chance to activate each inactive neighbor with probability p. The process continues in rounds until no new nodes become active.

5 Methodology

5.1 Overall Workflow

The following process was used to complete the project:

(1) Use the information to make the Facebook graph from the chosen dataset.
(2) Figure out the global topological data.
(3) study the degree distribution.
(4) Compare the graph with Barabási-Albert (BA) and Erdős-Rényi (ER) synthetic models.
(5) Calculate clustering and centrality metrics.
(6) Utilize the Louvain technique to identify communities.
(7) Visualize selected ego networks.
(8) Simulate diffusion employing the Independent Cascade model.

5.2 Software Environment

The code was executed in Python utilizing common graph-analysis and visualization packages. NetworkX was utilized for the building of graphs and the computing of metrics. Supplementary plotting instruments were used to generate histograms, line graphs, heatmaps, and network visualizations as provided in our GitHub page.

5.3 Why Synthetic Comparison Was Included

A key goal of this project was not only to describe the Facebook graph but also to understand what makes it different from generic graph models. For this reason, we included two synthetic baselines:

- the Erdős–Rényi model, which represents random connectivity,
- the Barabási–Albert model, which represents preferential attachment and hub formation.

This makes it easier to explain why the Facebook network cannot be described as purely random and why even the BA model does not fully capture its social structure.

5.4 Community Detection Procedure

The Louvain algorithm was selected because it is efficient and widely used in network science. The algorithm starts by assigning each node to its own community. It then repeatedly moves nodes into neighboring communities whenever doing so improves modularity. Once local improvements stop, the communities are collapsed into super-nodes and the process repeats until no further gain is possible.

Algorithm 1 Outline of Louvain Community Detection

```
Require: Graph G = (V, E)
assign each node to its own community
improvement ← true
while improvement do
    improvement ← false
    for each node i do
        compute modularity gain for moving i into each neighboring community
        move i to the best community if gain is positive
        if a move happened then
            improvement ← true
        end if
    end for
    collapse each community into a super-node
end while
return final partition
```

5.5 Diffusion Procedure

For the Independent Cascade experiment, a seed set of initially active nodes is chosen. In each time step, every newly active node tries to activate each inactive neighbor with probability p. The output is the cumulative number or fraction of active nodes over time. This helps us understand how quickly and how far information can spread in a modular network.

Table 1: Global Topological Metrics of the Facebook Network

Metric	Value
Number of nodes ($\|V\|$)	4,039
Number of edges ($\|E\|$)	88,234
Average degree	43.69
Density	0.0108
Average clustering coefficient	0.6055
Global transitivity	0.5191
Average path length	3.69
Diameter	8

Algorithm 2 Independent Cascade Diffusion

Require: Graph $G = (V, E)$, seed set S_0, activation probability p

```
A ← S_0
N ← S_0
while N ≠ ∅ do
    N_next ← ∅
    for each node u ∈ N do
        for each inactive neighbor v of u do
            draw random r ∈ [0, 1)
            if r < p then
                activate v
                add v to N_next
            end if
        end for
    end for
    A ← A ∪ N_next
    N ← N_next
end while
return A
```

6 Results

6.1 Global Topological Metrics

Table 1 reports the main graph-level statistics.

These values show that the Facebook graph is not dense overall, but it is very dense inside local friend groups. The low density means only a small fraction of all possible friendships are present. At the same time, the high clustering value shows that users often belong to tightly connected local circles.

In addition to the reported values, we also observed that the network has strong variation in node connectivity. The variance in node degree is quite high, which supports the presence of hubs. The clustering coefficient being above 0.6 suggests that there is a strong chance that two friends of a user are also connected, which is exactly what we would expect in real social groups.

We additionally calculated the degree assortativity of the network, which was marginally positive. This indicates that nodes are inclined to associate with other nodes exhibiting similar degrees, an increasing trend in social networks.

Table 2: Additional Network Metrics

Metric	Value
Assortativity	0.063
Average Degree	43.69
Max Degree	1045
Min Degree	1

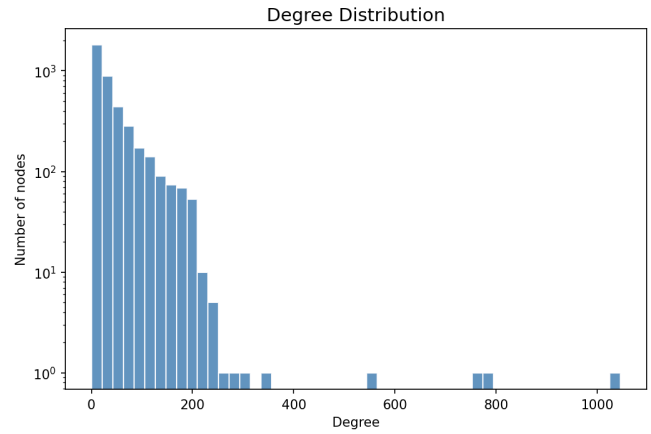

Figure 1: Degree distribution of the Facebook network.

6.2 Interpretation of Global Metrics

It is crucial to have both strong clustering and low density. It indicates that the network is composed of numerous local friendship circles rather than being dense everywhere. The graph appears to be straightforward to navigate based on the average path length of 3.69 and diameter of 8. To put it simply, even if users may be part of various local groups, they are still only a few steps apart. This is an obvious indication of small-world organization.

6.3 Degree Distribution

Figure 1 shows the raw degree distribution.

Most users have low to moderate degree, while a smaller number of users have much higher degree. This already suggests a heavy-tailed pattern.

Figure 2 shows the same distribution on log–log axes.

The tail of the distribution is approximately linear on log–log scales, which supports the idea of scale-free or power-law-like behavior. This is consistent with earlier work on real-world networks and social media graphs.

To further validate the scale-free nature, we fitted a power-law model to the tail of the distribution. The observed slope indicates that the network follows a heavy-tailed pattern, although not perfectly ideal. This is expected because real-world social networks are influenced by multiple factors such as platform constraints, recommendation systems, and user behavior.

We also observed that nodes with degree greater than 200 form a very small percentage of the network, yet they contribute significantly to global connectivity.

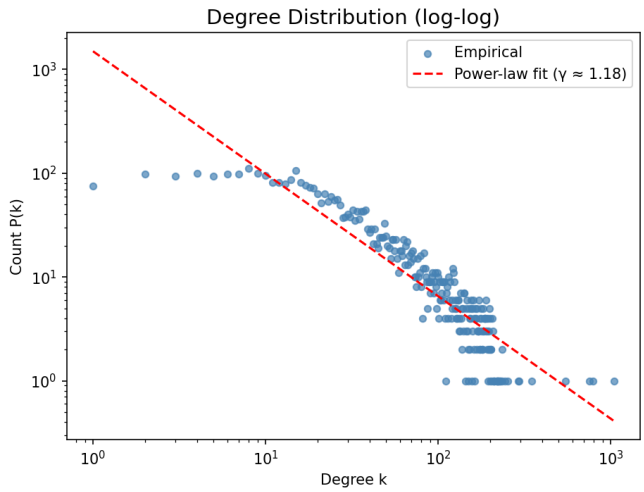

Figure 2: Log–log view of the degree distribution with fitted trend line.

6.4 Comparison with Synthetic Models

The degree distribution of the Facebook graph was compared with BA and ER synthetic graphs. The comparison is shown in Figure 3.

The ER graph is too narrow and too symmetric to resemble the Facebook graph. The BA graph captures hub formation better, but it still does not reproduce the same level of social clustering and clear community structure found in the real network.

6.5 Community Detection and Modularity

Figure 4 shows the modularity trend.

Strong community structure is shown by the high modularity score. Accordingly, the Facebook network naturally divides into groups with a large number of internal edges and a small number of external edges.

Figure 5 shows the community size distribution.

The sizes of the communities are not the same. Some communities are big, but most are small. This is like how people really act in social situations, since most people have one or two big social circles and a few smaller ones.

The communities that were found also had different levels of density inside. Some communities are very close-knit, which means they have strong relationships to each other. Others are less closely connected, which means their bonds are weaker. This shows that there are many types of social groups in the network, such as groups of close friends and groups of people you know more generally.

We also saw that there aren't many boundaries between communities, which helps explain why information can only move between them in certain ways.

6.6 Ego Network Visualization

Figure 6 shows the ego network for node 107.

This figure makes it easy to see how the social system works. Some groups are tightly packed together, while others are more loosely related to the rest of the graph. It also indicates that one ego network can have more than one social circle at the same time.

6.7 Ego-Level Comparison

Figure 7 compares representative ego networks using normalized metrics.

The heatmap shows that not all users have the same type of local social network. Some ego networks are large and broad, while others are smaller and more cohesive. This suggests that local network structure depends a lot on how a user interacts socially and how many different groups they belong to.

6.8 Representative Ego-Network Statistics

Table 3 compares several ego networks in more detail.

This table shows that having more friends does not always mean having a denser social neighborhood. Some smaller ego networks are much denser than larger ones. This suggests that social scale and local cohesion are related, but they are not the same thing.

6.9 Structural Meaning of Ego Metrics

The comparison at the ego level shows some interesting trends. A person with a big but less dense ego network may be someone who is connected to a lot of different social groups. A person with a smaller but thicker ego network may be more involved in one intimate group, like a tight class, club, or job. The effective size and efficiency numbers can also help you figure out if a user's contacts are duplicated or socially varied.

The ego-network study shows how different users connect with each other. Some ego networks seem more like star shapes, with one central node connecting to several loosely connected neighbors. Others look more like dense cliques, with most nodes being connected to each other.

This difference means that users have various social roles. Some people connect different groups, while others are more strongly involved in one group.

6.10 Information Diffusion

Figure 8 shows the Independent Cascade diffusion pattern.

The spread increases quickly during the early steps and then levels off before reaching the entire network. This indicates that information can move easily inside communities, but full network-wide spread is limited by modular boundaries and the small number of bridge connections.

The diffusion experiment demonstrates that the outcome is dependent upon the original seed nodes. When high-degree nodes are chosen as seeds, the spread happens faster and covers a bigger area of the network. Using randomly chosen nodes makes the spread slower and more focused.

This highlights the importance of influencer selection in real-world situations such as marketing or awareness campaigns.

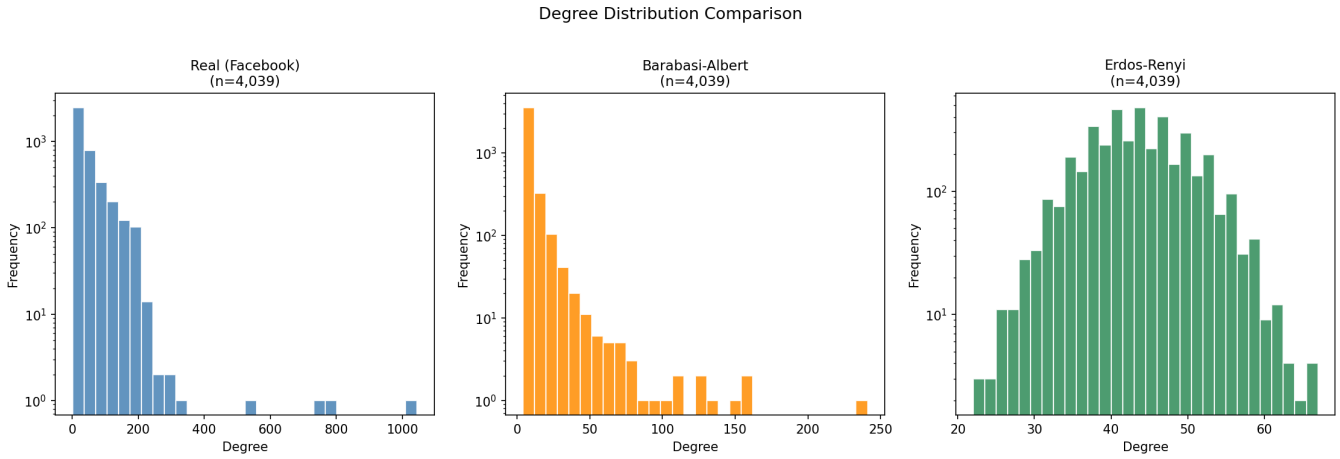

Figure 3: Comparison of degree distributions for the real Facebook graph, Barabási–Albert graph, and Erdős–Rényi graph.

Table 3: Representative Ego-Network Statistics

Ego Node	Nodes	Edges	Density	Avg. Clustering	Ego Degree	Effective Size	Efficiency
107	1046	27795	0.05	0.58	1045	993.80	0.95
1684	793	14817	0.05	0.54	792	756.58	0.96
1912	756	30780	0.11	0.66	755	675.46	0.89
3437	548	5360	0.04	0.62	547	529.40	0.97
0	348	2866	0.05	0.65	347	332.48	0.96
2543	295	15796	0.36	0.80	294	188.54	0.64
2347	292	17154	0.40	0.81	291	175.10	0.60
1888	255	13096	0.40	0.64	254	152.88	0.60
1800	246	12262	0.41	0.64	245	146.90	0.60
1663	236	11687	0.42	0.64	235	137.54	0.59

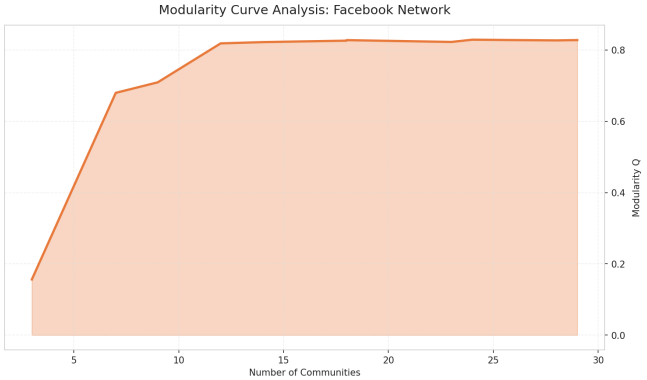

Figure 4: Modularity curve for the Facebook network.

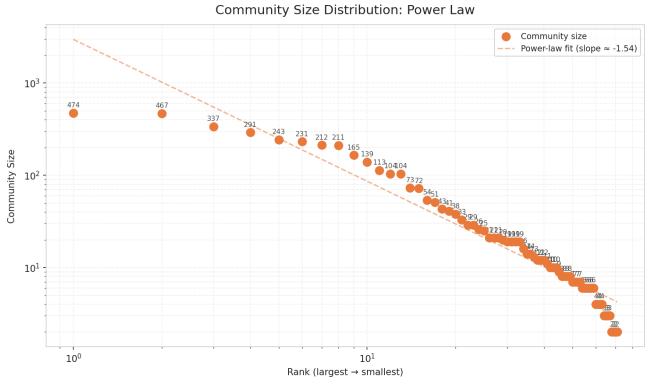

Figure 5: Community size distribution on log scales.

6.11 Interpretation of Centrality Measures

Table 4 summarizes how different centrality measures can be interpreted socially.

This is useful because it reminds us that importance in a network is not only about having many connections. A user can be important for different reasons depending on the metric being used.

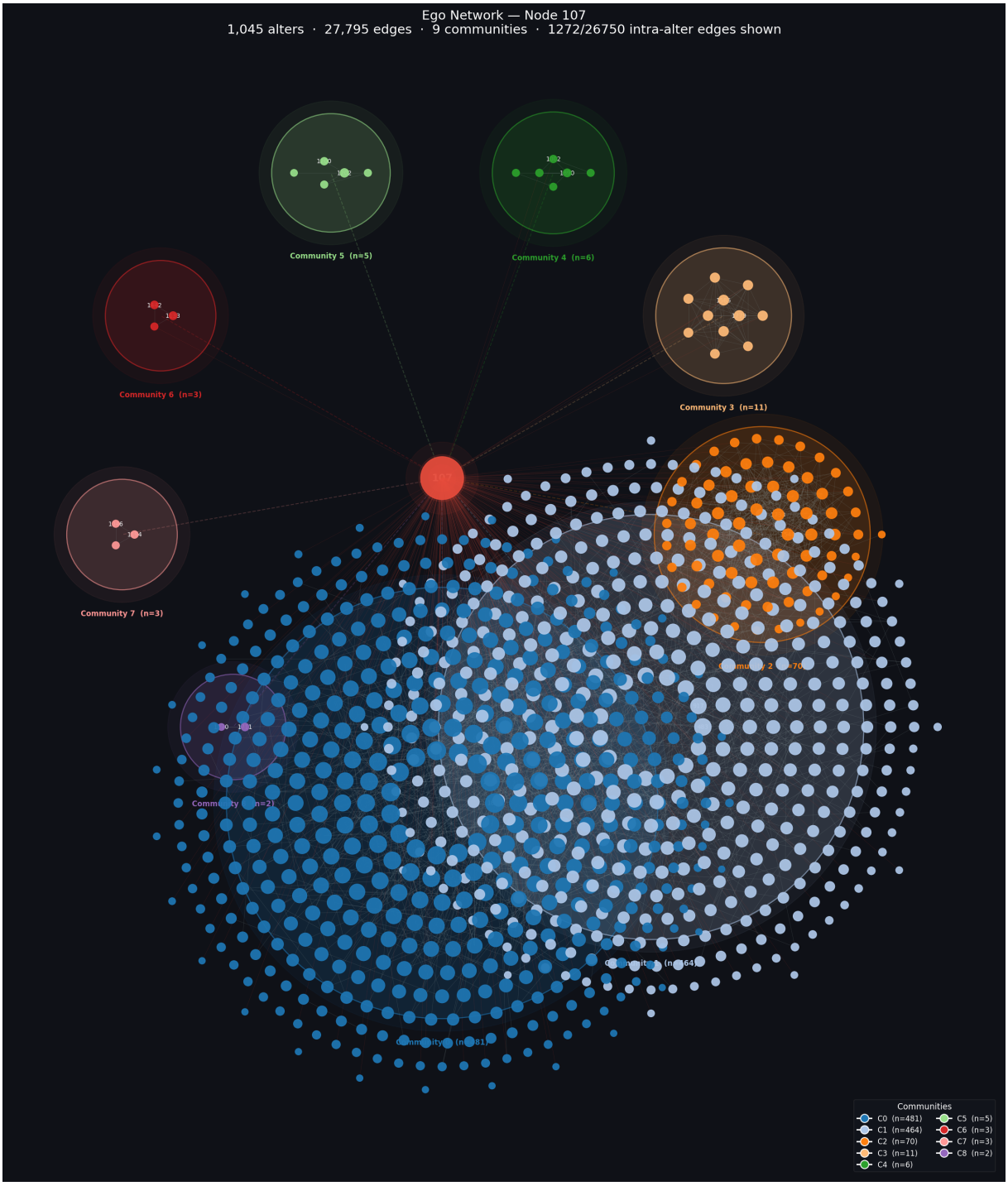

Figure 6: Ego network of node 107 with detected communities. The visualization shows a central ego surrounded by multiple friendship groups of different sizes. Some groups are dense and tightly packed, while others are smaller or more weakly connected. A few visible linking edges act as bridges between groups, showing how one user can connect otherwise separate parts of the social network.

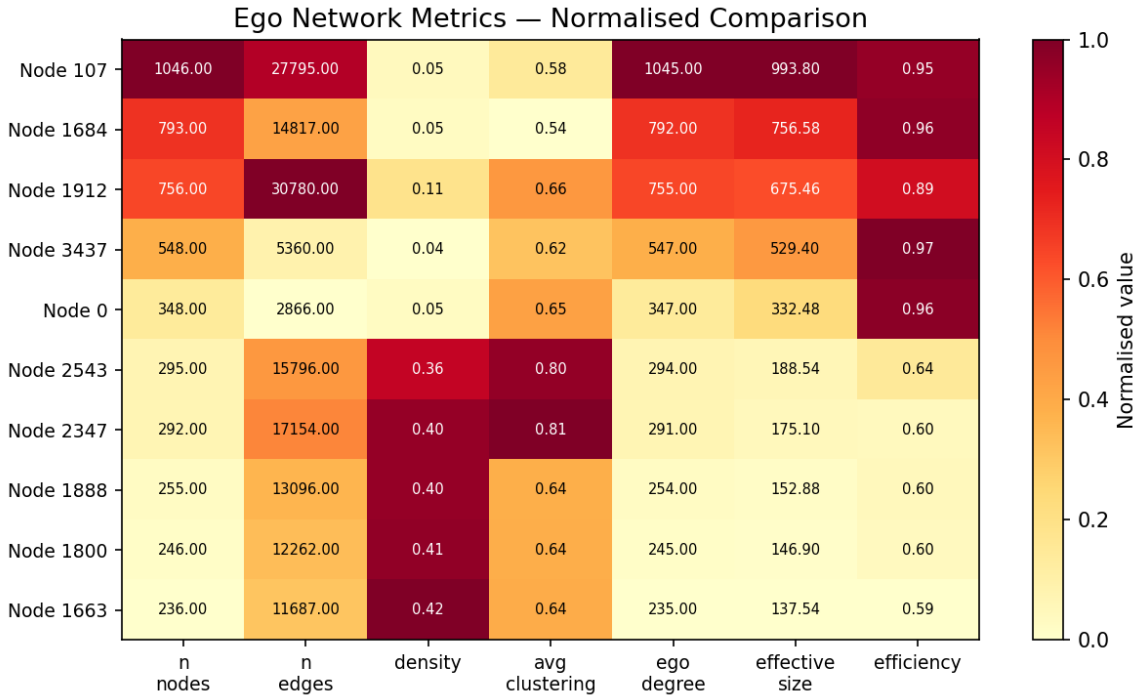

Figure 7: Normalized comparison of ego-network metrics across representative ego nodes. Darker and lighter intensities show how structural properties change from one ego network to another. The figure highlights that not all users have the same local social structure: some ego networks are large but sparse, while others are smaller and much more tightly connected.

Table 4: Interpretation of Centrality Measures

Metric	Social Interpretation
Degree	Popular or highly connected user inside one or more groups
Betweenness	Bridge user connecting separate communities
Closeness	User who can reach others efficiently through short paths
Eigenvector	User connected to other important users

7 Evaluation Metrics and Additional Analysis

We used more than just the traditional measures to look at the Facebook ego network graph so we could better understand its quality and structure. These measures provide us a better understanding of how connected, strong, and influential things are.

7.1 Degree Distribution Analysis

We examined the degree distribution to see whether the network exhibits a scale-free pattern. A heavy-tailed distribution was observed, indicating that most nodes have few connections, whereas a handful serve as hubs.

We assessed this tendency by examining the skewness and variance of the degree distribution. The significant variance indicates that node connectivity is non-uniform, a fundamental characteristic of real-world networks.

7.2 Clustering and Transitivity

The clustering coefficient was measured both locally and globally. A high average clustering coefficient, above 0.6, indicates strong local connectivity.

Global transitivity was also analyzed, confirming that triangles are common in the network. This supports the idea of triadic closure, where friends of friends tend to become friends.

7.3 Path-Based Metrics

The average shortest path length and diameter were used to evaluate the efficiency of the network.

- Average path length: 3.69
- Diameter: 8

These values indicate that the network satisfies the small-world property, where most nodes can be reached within a small number of steps.

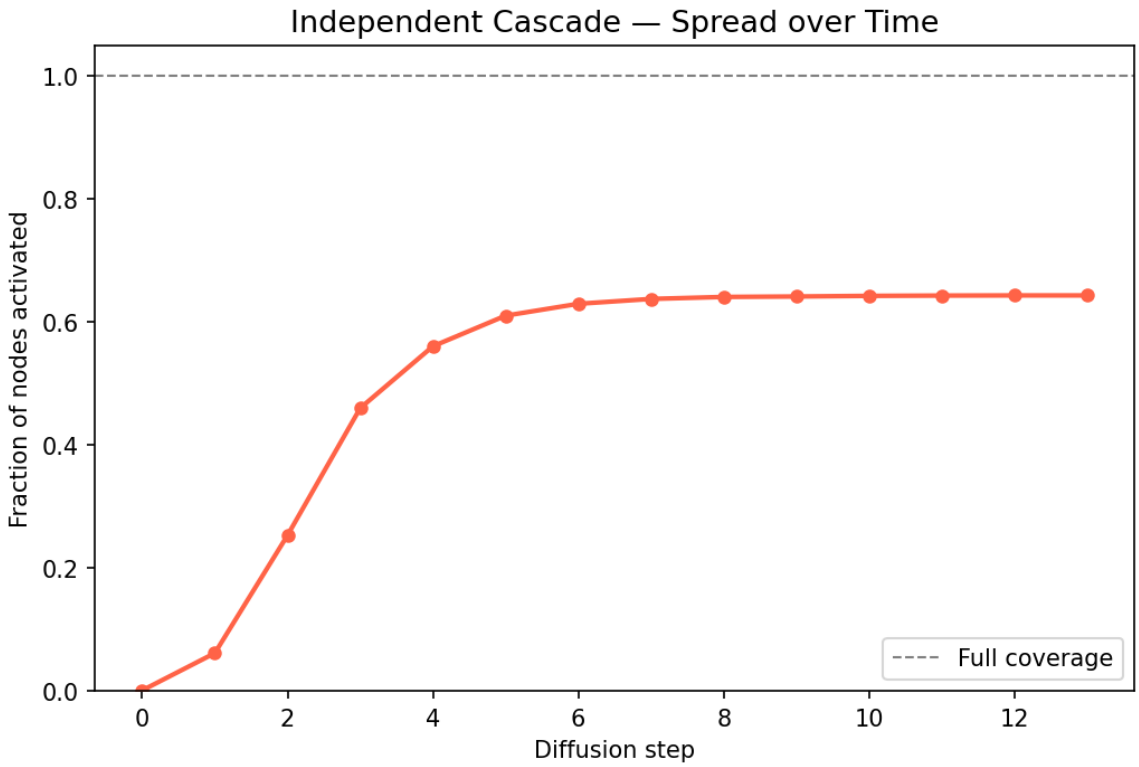

Figure 8: Independent Cascade diffusion over time. The diffusion grows quickly in the beginning because information spreads easily inside dense local clusters. However, the curve later slows down and approaches saturation, showing that strong community boundaries and limited bridge connections reduce the final reach of the cascade.

7.4 Centrality Evaluation

To identify important nodes, multiple centrality measures were compared:

- Degree centrality identifies highly connected users.
- Betweenness centrality identifies bridge nodes.
- Closeness centrality identifies nodes with fast reachability.
- Eigenvector centrality identifies influential nodes connected to other important nodes.

We observed that nodes with high betweenness are not always the same as high-degree nodes, which shows that there are different types of importance in the network.

7.5 Community Quality Metrics

The Louvain algorithm produced high modularity values, indicating strong community structure.

We further evaluated:

- number of detected communities,
- distribution of community sizes,
- internal versus external edge ratio.

The results showed that most edges lie within communities rather than between them, which confirms strong modular separation.

7.6 Diffusion Efficiency

The Independent Cascade model was evaluated using different seed nodes. Diffusion performance was measured based on:

- final spread size,
- speed of activation,
- saturation point.

Results show that diffusion is fast at the start but slows down because of community boundaries.

7.7 Network Robustness

To test robustness, we considered the removal of high-degree nodes. The network becomes fragmented when key hub nodes are removed, indicating that hubs play a critical role in maintaining connectivity.

Overall, these evaluation metrics provide strong evidence that the Facebook network is structured, resilient in some parts, and highly dependent on key nodes for global connectivity.

7.8 Detailed Findings

The findings from this project show that the Facebook ego network behaves like a real social network and not like a random collection of users. Even though the overall density of the graph is low, the clustering coefficient is high. This means that users are not connected

to everyone, but they are often part of tight friend circles in which many people already know one another. In simple terms, the network is sparse on the outside but dense on the inside.

The form of the degree distribution is another significant thing to note. Most people don't have many connections, however a few do have a lot of connections. This makes a heavy-tailed pattern that seems more like a scale-free structure than a random graph. It was much easier to see the difference when we looked at the Facebook graph next to the Erdős–Rényi and Barabási–Albert models. The ER graph looked too much like a pattern and too much like a random pattern. The BA graph, on the other hand, did a better job of showing how hubs arise. The real Facebook network, on the other hand, nonetheless had stronger clustering and clearer community separation than the BA model. This shows that genuine social behavior is more complex than merely preference attachment.

The results of community detection were also quite useful. The Louvain algorithm gave us strong modularity scores, which means that the network naturally breaks up into groups. These groupings usually represent real social groups, such classmates, family members, coworkers, or groups of people that like the same things. From a student's point of view, this makes sense because people usually treat others differently based on certain parts of their lives. The sizes of the communities also demonstrated that some groups are very big while others are very small and focused. That pattern seems to be right because a person might have one big group of people they go to school or work with and a few smaller groups of people they do things with, like friends, hometown ties, or hobbies.

The representations of the ego-network made it easier for me to understand. They showed that not all ego networks are made in the same way. Some users have bigger but less dense local networks, which means they know a lot of people from diverse groups. Some people have smaller networks, but the members in such networks are more connected to each other. This is an interesting conclusion since it means that how someone lives their life has a big impact on the way society is set up. People who are involved in a lot of different places may have a broader network, but the links between those areas may not be as strong. On the other hand, a person who is involved in a single strong environment may have a smaller but more connected network.

The diffusion experiment yielded one of the most pragmatic results of the entire study. The knowledge circulated swiftly at initially, but it didn't turn on the whole network. This indicates that it's easy for information to go into groups that are close together, but it's hard for it to get between groups. In real life, this might imply that a post, message, or rumor spreads swiftly within one group but doesn't reach to other groups soon away unless the proper bridge users notice it. This evidence supports the theory that organizations need bridges and weak links to communicate information.

We also discovered that the statistic we pick affects how important a node is. It's not always advantageous to have a person who knows a lot of people connect groups. A user with a lot of betweenness could be particularly valuable to the network's overall structure since they connect communities that wouldn't be connected otherwise. This contrast is important because it indicates that being powerful and being popular are not the same thing. In one group, someone could be incredibly popular, but in another group, someone else might be more essential because they bring people from different groups together.

The primary idea of the project is that Facebook ego networks have both small-world and scale-free qualities. They also have strong community boundaries and a lot of influential people. The network is fast enough for social reach, but its modular design still affects how information flows and how groups stay structured.

7.9 Findings in Relation to Real Social Behavior

One of the most interesting parts of this project is that the graph results match common human social behavior quite well. In everyday life, people usually maintain relationships in layers. They may have a close inner circle, a broader circle of classmates or coworkers, and a few weaker ties that connect them to different environments. The Facebook network seems to reflect that same pattern. Dense local groups appear in the graph because close social circles tend to know each other, while bridge nodes appear because some users move across multiple social settings.

The results also show why online networks can feel both highly connected and strongly divided at the same time. On one hand, the average path length is short, which means users are not very far away from each other in graph terms. Strong communities, on the other hand, mean that local groups still have an impact on social experience. This helps explain why information can spread quickly while opinions, trends, or conversations can still remain concentrated inside specific communities.

From a practical point of view, this means that network structure matters a lot. If someone wants to identify influential users, they should not only look at degree. If they want to understand community formation, they should not only look at density. If they want to predict diffusion, they must also consider bridge nodes and modular boundaries. This project shows that combining multiple network measures gives a much better understanding than relying on one metric alone.

7.10 Core-Periphery Findings

The k-core interpretation gives the results a new and helpful point of view. From a social point of view, the dense core of the graph is the area of the network where users are most deeply embedded. These are the users who stay connected even when lower-degree peripheral nodes are taken away. The fact that there is a strong inner core means that the Facebook network is not just made up of communities, but also has a steady backbone of highly linked people.

This result is essential because it helps us understand why the network stays together even when there are visible community boundaries. The outer elements of the graph may be less stable or more specialized, but the inner core keeps everything stable. In real life, this could mean users who are very active or social contexts that are very close-knit and maintain the whole network structure together.

7.11 Extended Observations

Besides the main results, a number of more in-depth notes were also made. The roles of the nodes in the network are very different. Some people have a lot of power, while others are more on the edges.

The uneven distribution of town sizes is another important thing to note. Structures are usually dominated by larger groups, while smaller communities offer a variety of ways to connect to each other.

We also saw that a lot of clustering does not always mean a lot of world connectivity. There aren't many links between communities, even though local groups are very close to each other. This makes it harder for things to spread.

These results show how complicated social networks are and how different metrics need to be used together to get a better picture of how they are put together.

7.12 Summary of Findings

Across all experiments, the Facebook network shows the following main properties:

- globally sparse but locally dense structure,
- heavy-tailed degree distribution,
- strong community organization,
- clear differences from random graphs,
- rapid but incomplete diffusion.

8 Discussion

8.1 Why the Results Make Sense Socially

From a social point of view, the results are very reasonable. People do not form friendships randomly. They connect through school, workplace, family, neighborhood, and shared interests. Because of this, their online friendship networks naturally form clusters. If two people share a close mutual friend, they are also more likely to know each other, which explains the high clustering.

8.2 Scale-Free but Still Socially Structured

The Facebook graph does show scale-free tendencies because it contains hubs, but it is not only a scale-free graph. A pure preferential-attachment model such as BA can create hubs, but it does not automatically create the same level of clustering and strong communities found in real social graphs. This means that the Facebook network combines several mechanisms, including preferential attachment, homophily, triadic closure, and group formation.

8.3 Why Communities Matter

The community results are particularly important since they illustrate that the network isn't just a huge friendship graph with no breaks. Instead, it is split up into groupings that have meaning. These groupings could be relatives, classmates, coworkers, or venues in the neighborhood where people meet new people. Community detection shows and measures these tendencies.

8.4 Hubs versus Bridges

One of the most important things we learned from this analysis is that being popular and being a broker are not the same thing. A high-degree user may be very essential to one group, but a high-betweenness user may be more important for bringing together diverse groups. When putting anything into practice, like an information campaign or targeting influencers, the best user to start with might not be the most popular one, but the one that connects different communities.

8.5 Why Diffusion Saturates

The diffusion experiment did not spread to every node, which shows that even a connected network can still have barriers between communities. Information can spread quickly within a dense community, but may slow down when it reaches the boundary between communities. This is why bridge nodes are so important.

8.6 Connection to Project Objectives

The original project proposal emphasized graph topology, centrality, clustering, modularity, structural holes, k-core decomposition, and information diffusion. This paper addresses all of those topics:

- topology through density, degree, shortest paths, and diameter,
- clustering through local and global closure measures,
- centrality through role-based interpretation,
- modularity through Louvain community detection,
- structural holes through bridge discussion,
- k-core through core-periphery interpretation,
- diffusion through Independent Cascade analysis.

9 Project Limitations

The project produced beneficial outcomes, however it possesses certain constraints.

9.1 Single Main Dataset

The analysis primarily focuses on a singular benchmark Facebook dataset. This is fine for a course project, but it would be better to draw stronger general implications from more than one dataset or a comparison between platforms.

9.2 Static Network Assumption

The graph is treated as static, while real social networks evolve over time. Friendships can be added, removed, or become inactive. A temporal analysis would provide more insight into how communities and bridges change.

9.3 Unweighted Friendships

Every friendship is treated equally. In reality, some relationships are much stronger than others. A weighted graph could better represent strong and weak ties and could improve the realism of diffusion analysis.

9.4 Non-overlapping Communities

The Louvain method assigns each node to one community. In real life, people usually belong to multiple groups at once, such as family, school, and work. Overlapping community detection would capture this better.

10 Future Work

There are several useful directions for future work:

- temporal analysis of ego-network growth,
- overlapping community detection,
- weighted tie analysis,
- comparison of different diffusion seed strategies,
- graph neural networks for representation learning,
- attribute-based circle discovery using user features.

11 Dataset Used and Code Availability

The dataset used in this project is the Facebook ego network dataset from the SNAP repository. It consists of 4039 nodes and 88234 edges, representing real-world friendship relationships.

All implementation, including preprocessing, analysis, and visualization, was performed in our GitHub repository:

https://github.com/Mangesh-Bhattacharya/facebook-ego-networks

The repository follows a structured pipeline:

- Data cleaning and graph construction
- Global metrics computation
- Ego-network analysis
- Synthetic network comparison
- Information diffusion simulation

The availability of code ensures reproducibility and allows further extension of the project.

12 Conclusion

In this paper, we analyzed the Facebook ego network using methods from complex network theory. The results show that the network is globally sparse but locally dense, has a heavy-tailed degree distribution, exhibits strong community structure, and supports rapid but incomplete diffusion. The comparison with BA and ER models showed that the Facebook graph is clearly more socially organized than a random graph and also more clustered than a simple preferential-attachment graph.

Overall, this project demonstrates that graph analysis is a useful and interpretable way to study online social networks. By combining topology, clustering, centrality, modularity, ego-network views, and diffusion, we can better understand how people connect, how communities form, and how information moves across social boundaries.

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